

Scintillations



सोलव
PET CT



**SPECT
LAB**

nuclear medicine services



The fourth Radiation safety committee meeting of spect-lab was held on 3rd august 2008. Members of the committee along with invited guests are seen in following order from left to right Ms. Sarika, Ms. Depanjali , Sri. Ranjeet Mahajan, Sri. Sailesh, Dr. Dhanpati, Sri. Shankar Magar, Dr. Rajshekhar Rao, Sri Pradeep Kumar, Dr. P. S. Soni, Dr. B. S. Gill, Dr. Pallavi Solav, Mrs. Dhanpati.

Introduction

Last month I visited the nuclear medicine department of Yonsei University's Severance hospital in Seoul. This happens to be one of the largest hospitals in Seoul and the nuclear medicine department (which has seven gamma cameras and four PET scans) is one of the busiest departments. They perform on an average 50 to 60 PET-CT scans every day and about 180 general nuclear medicine studies.

Prof JD Lee Head of the Nuclear medicine was kind to allow me to be an observer for a week. Prof Lee's interest is related to Brain receptor imaging

Prof M Yun with whom I spent most of the time reading PET _CTs was a host defacto. Prof Yun happens to be extremely systematic in PET-CT interpretation.

There were other faculty members and residents in the hospital whom I found very hospitable, helpful and polite. Dr Cho, Dr Park, Dr Seo and Mr Park the radiation physicist were all very helpful in some way or the other during my visit to their department.

The weather in Seoul was perfect with highest temperature touching 25 degreeC. This was considered to be pretty hot by the local population.

Most of the communication is in local Korean language but scientific deliberations are with the help of English. The reports are also made in English. People on the street are quite helpful and the place is safe.

Forty years since independence, Korea has achieved remarkable growth in terms of development. They are indeed a progressive lot; they don't despise the tax payer (the current President happens to be the ex-chief of Hyundai). Contrast this to our state of affairs, a so called independent country, where people have freedom to spit where they like, void at their instinct, defecate at their place of choice. Sixty years into independence the tax payer is still looked down upon, babudom flourishes like nowhere else in the world and the right to abuse your job security is exploited to its summit. Truly speaking there is a vertical divide, not between marathi speaking and non marathi speaking, but between efficient system versus inefficient system, not between the rich and the poor, but between the performers and non performers, not between the public and private sector but between redundant policies/laws and the desire to change them. Hope that there will be a turnaround in coming years...

Now, lets look into some interesting PET-CT and general nuclear medicine studies.



Shrikant Solav

Case 4. *Hibernating myocardium*

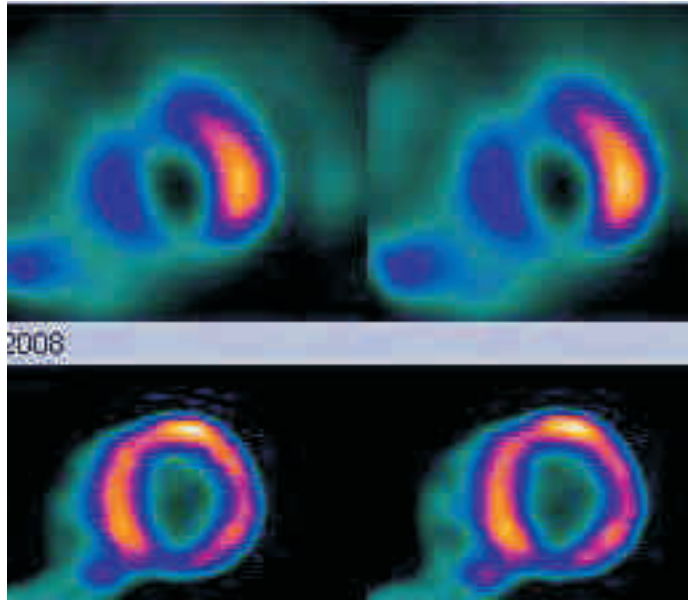


Fig.4

Hibernating myocardium is defined as chronic ischemic left ventricular dysfunction which is potentially capable of improving in function upon successful revascularization. Thus the gold standard for hibernating myocardium is documentation of improvement in left ventricular function post revascularization.

This entity needs to be differentiated from scar related left ventricular dysfunction (which will not improve in function post revascularization and also carries high peri-operative mortality.)

It also needs to be differentiated from dilated cardiomyopathy, a condition in which the myocardium itself becomes weak in absence of coronary stenosis.

Chronic ischemic left ventricular dysfunction is a challenging entity in cardiology. It carries high long term mortality rate if not revascularised.

There are three important parameters on the basis of which one can identify hibernating myocardium.

1. Perfusion status. (Thallium scan)
2. Functional reserve (low dose dobutamine gated spect/ echo study)
3. Metabolic reserve (FDG-PET scan)

If a segment of myocardium is dysfunctional as identified on echocardiogram, a simple way to look for viability is to check for perfusion status of the respective segment. An akinetic segment is considered to be viable if the perfusion of such a region is preserved. The perfusion in such segments is not expected to be normal, but if the quantity of tracer uptake remains more than 35% of the non-infarcted territory, it is considered to be a reasonably good indicator of myocardial viability.

Identification of functional reserve of the myocardium is used as a parameter of viability when the perfusion is preserved in the face of left ventricular dysfunction to distinguish between ischemic cardiomyopathy and dilated cardiomyopathy. Using dobutamine challenge gated spect study a dual mode response is a highly specific marker of hibernating myocardium. (low left ventricular ejection fraction which improves in response to low dose infusion of dobutamine and which shows

deterioration in function as the rate of infusion is increased to higher level)

The third important parameter is that of metabolic reserve. A dysfunctional segment which is devoid of perfusion (on Thallium scan) shows preserved metabolism on FDG-PET scan. This is called as perfusion-metabolism mismatch.

Normal myocardium under fasting state preferentially uses fatty acids for its metabolism. In postprandial state there is a shift of metabolism to glucose.

A dysfunctional myocardial segment that is devoid of perfusion will continue to show glucose uptake in postprandial state as shown in the figure 4. (This is called as perfusion- metabolism mismatch)

Case 5. Assessing the efficacy of chemotherapy

An important aspect of PET-CT imaging is its objective value in determining the efficacy of chemotherapy.

This case highlights the importance of PET scan over and above that of conventional imaging modality such as CT scan in assessing the treatment response of tumor.

The lady 45 years of age, presented with chest pain and weight loss. There was a mass seen radiologically on CT scan. PET scan demonstrated intense metabolic activity in this lesion with involvement of ipsilateral hilar lymph node. (Fig 5A) Exploration revealed the mass to be adherent to the myocardium and hence could not be removed.

She was subjected to chemotherapy. Repeat PET scan following 4 cycles of chemotherapy revealed complete resolution of metabolic activity of the tumor. The corresponding CT scan revealed a small area of collapse-consolidation which was devoid of FDG uptake. (Fig 5B) The myocardium shows physiologic FDG uptake in the post therapy study, which is not so obvious in the pre-treatment scan in view of intensely hyper-metabolic adjacent tumor. In view of absence of tumor activity subsequent radiation was spared.

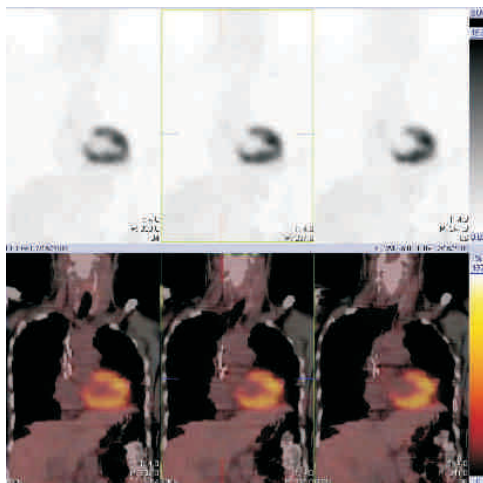


Fig.5A

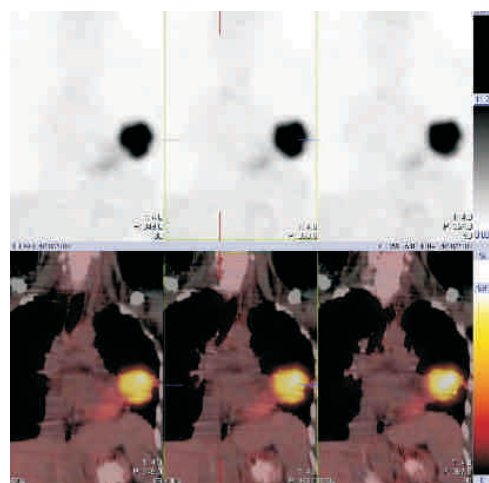


Fig.5B

Case 6. Early metabolic response to chemotherapy: (PET Vs CT)

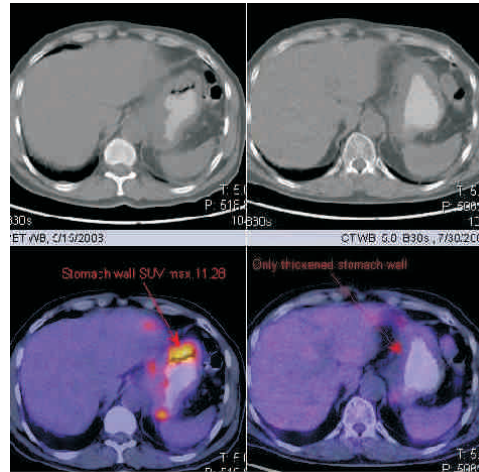


Fig. 6

This is a 63 year old male patient who was diagnosed as adenocarcinoma of stomach. The pre chemotherapy CT scan of chest and abdomen revealed thickened stomach wall(body and antrum) with multiple lymph nodes involving celiac axis, precaval, retrocaval, multiple retroperitoneal, left supraclavicular and left cervical level IV nodes.

FDG PET-CT performed at the same time revealed metabolically active lesions at the same sites. (Figure 6 left panel)

Patient received 3 cycles of chemotherapy. Follow up CT of chest & abdomen and FDG PET-CT were performed to look for chemotherapy response.

The post chemotherapy CT scan was status quo with no change in the structure of stomach wall as compared to the pre-chemotherapy scan. (Figure 6 right panel)

However the FDG PET-CT revealed complete resolution of FDG uptake (metabolic activity) at the primary site in the stomach and immediate regional nodes like peri-gastric & celiac axis nodes. There was partial response in rest of the lesions.

This case highlights the importance of FDG PET in assessing tumour response as against conventional imaging modalities such as CT. The former shows metabolic activity of the tumour that is more accurate in showing the efficacy of chemotherapy. In contrast to PET scan, the CT scan depicts structural changes that do not alter early in the post therapy period.

Case 7.



Fig. 7C

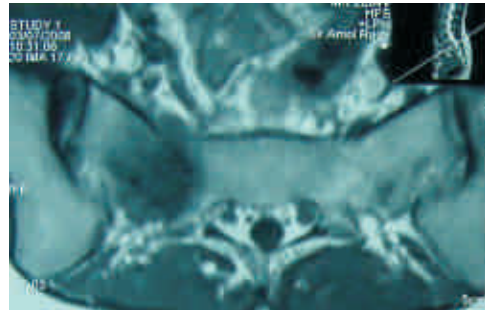


Fig. 7B

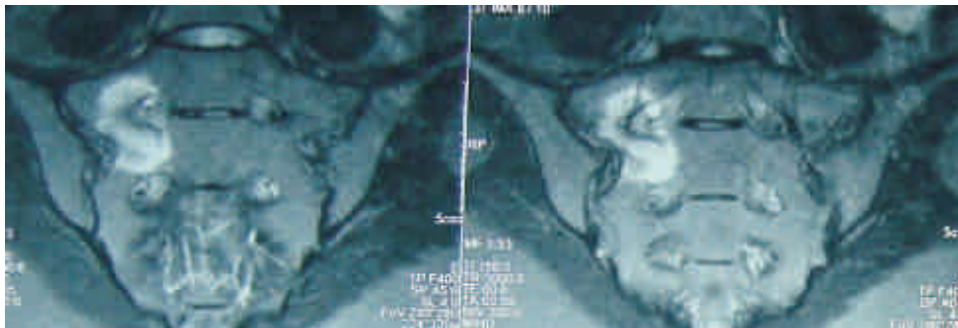


Fig. 7A

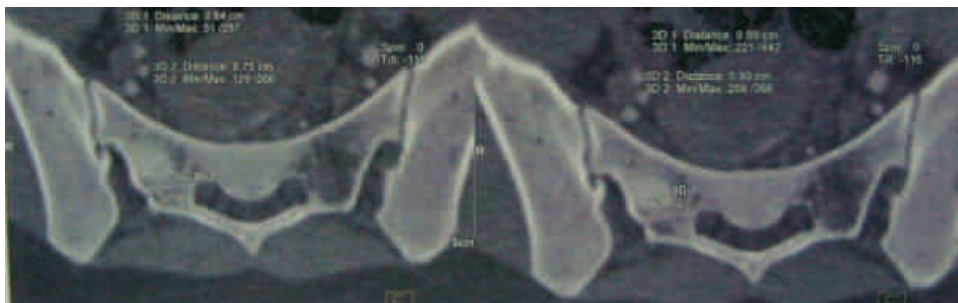


Fig. 7D

This 19 years old lady presented with pain in the region of right hip and low backache. MRI revealed marrow oedema at the level of S1/2 vertebra. (Fig 7A,B)

Bone scan revealed well defined hot spot in the posterior element of S1 vertebra towards its right lateral aspect. (Fig 7C) This was quite suggestive of osteoid osteoma.

Later on CT scan revealed osteolytic lesion with central nidus in the same region. (Fig 7D) The lesion was eventually excised.

Interestingly there was a method adopted to confirm the excision of the tumor. The patient was injected with radioisotope, a few hours prior to the surgery. Per operative bits of bone tissue were collected in saline medium and sent to the nuclear scan to be imaged under the gamma camera. An assumption was made that the piece of bone harboring the nidus would appear distinctly hot.

Histology confirmed the diagnosis of osteoid osteoma.

Scinti - Info

- *Skepticism is the longest word that one can make using alternate hand while typing.
- *No piece of paper can be folded more than seven times
- *The shortest complete sentence in the english language is “ Go “.
- *A chameleons tongue is twice as long as its body.
- *“ I never remember feeling tired by work though idleness exhausts me completely.
- Sherlock Holmes.

List of Investigations

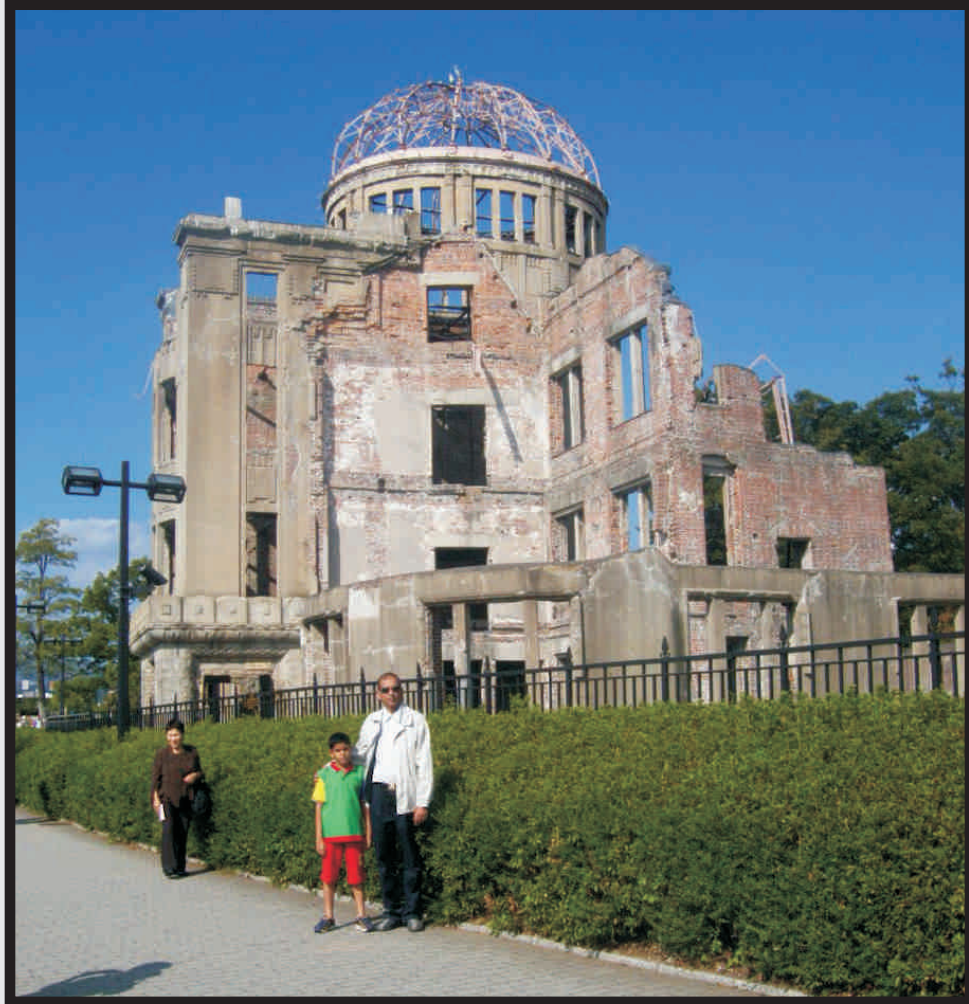
- * WHOLE BODY PET-CT
- * CARDIAC PET

- * NEUROLOGY BRAIN PET-CT
- * FLUORINE : BONE SCAN-CT

Conventional tests

- | | |
|-------------------------------------|---|
| 1) THYROID SCAN | 15) G I BLEED STUDY / RBC TAGGED SCAN |
| 2) RADIOIODINE ABLATION | 16) PUO |
| 3) STRESS THALLIUM SCAN | 17) CSF CISTERNOGRAPHY
(NORMAL PRESSURE HYDROCEPHALUS) |
| 4) CAPTOPRIL RENAL SCAN | 18) BRAIN SPECT(EPILEPSY / ALZEIMERS) |
| 5) DTPA RENAL | 19) GASTRIC EMPTYING |
| 6) EC RENAL | 20) G. E. REFLUX/MILK SCAN |
| 7) DMSA RENAL | 21) TESTICULAR SCAN FOR TORSION |
| 8) VUR STUDY | 22) MIBG PHEOCROMOCYTOMA) |
| 9) BONE SCAN | 23) P-32 THERAPY OF SKELETAL METASTASIS |
| 10) GALLIUM SCAN | 24) MUGA |
| 11) LUNG VENTILATION/PERFUSION(V/Q) | 25) BONE SPECT |
| 12) LIVER COLLOID SCAN | 26) PARATHYROID SCAN |
| 13) HEPATOBILLIARY SCAN(HIDA) | 27) IODINE SCAN |
| 14) MECKELS SCAN | 28) DACRYOSCINTIGRAPHY |

- * If you have any interesting case in which nuclear medicine made an impact, please send the details so that it may be published in the forthcoming issues.



Remains of the devastation caused by “Little boy” the uranium loaded weapon. Sixth of August reminds us of the atomic bombing of Hiroshima in the year 1945. At 8 am local time, an area admeasuring 1.6 km of this city was instantly charred killing as many as 180000 people spontaneously. Similar scene was witnessed on the 9th of August 1945 at Nagasaki. There goes an anonymous saying-

To smash the little atom
All mankind was intent
Now every day
The atom may
Return the compliment.

In nuclear medicine, you have an opportunity to use the atom in unraveling the illness that may remain inaccessible by other means.



The optimist sees opportunity in every danger:
the pessimist sees danger in every opportunity.
- **Winston Churchill**

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